

LESSON 1.7

DIVISION OF WHOLE NUMBERS



LESSON INTRODUCTION

MA.5.NSO.2.2 Divide multi-digit whole numbers, up to five digits by two digits, including using a standard algorithm with procedural fluency. Represent remainders as fractions.

LEARNING TARGETS

- I can find whole-number quotients of whole numbers.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency

ESSENTIAL QUESTIONS

- What are the different approaches or strategies to dividing whole numbers?
- What is the same or different about the partial quotient and long division methods?
- What is a remainder in the context of a long division problem?
- How are division and subtraction related?

LESSON NARRATIVE

In this lesson, students learn how to partition a model of a dividend into equal groups and relate the model to the partial quotient and long division methods of division for whole-number quotients and quotients with remainders. Using these approaches and two- and three-digit divisors up to five-digit dividends, students choose a preferred or most effective method to practice division and connect division to repeated subtraction.

COMPONENT SUMMARY

1.7.1 WARM-UP (~5 MIN.)

What is Math Anxiety video and reflection

1.7.3 GUIDED INSTRUCTION (10 MIN.)

Exploring partial quotient approach and connecting to a long division algorithm

1.7.5 YOUR TURN! (10 MIN.)

Students choose any method to solve division problems

1.7.2 GUIDED INSTRUCTION (10-15 MIN.)

Students use a model with base ten blocks to partition equal groups and divide with remainders

1.7.4 TRY IT (10-15 MIN.)

Partial products and long division practice with four-digit dividend by a two-digit divisor

1.7.6 WRAP-UP (~5 MIN.)

Growth Mindset reflection

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Partition
- Divisor, Dividend
- Quotient

MATERIALS

- 1.7.1 Warm-Up video: What is Math Anxiety
- (Optional) Base ten blocks

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand the concept of division as repeated subtraction.
- Students may confuse how to use the partial quotient method and the standard algorithm.
- Students may not recognize which values are the divisor and which values are the dividend.
- Students may have difficulty identifying the operation that is required in a real-world problem.

THINGS TO CONSIDER

- Position students to understand what the remainder represents in context and how to write it using the divisor as the denominator of the fractional remainder.
- Using the tables in the 1.7.2 and 1.7.3 Guided Instruction sections to identify multiples of the divisor can help students determine which values result in the smallest number of steps as well as the reasonableness of their solutions.
- This lesson contains rich vocabulary related to division; be sure to model and use the words correctly within context.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

To capture students' oral words and thinking when approaching the division problems in this lesson, consider collecting responses throughout the lesson and displaying them in a prominent space in the classroom. This routine can help students think or write about a response and refine and clarify those responses through conversation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

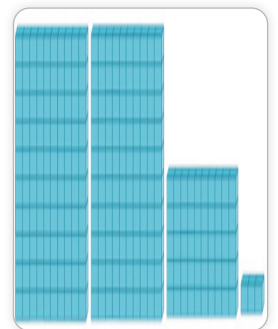
MA.K12.MTR.1.1 Participation in Effortful Learning

1.7.1 Warm-Up and 1.7.6 Wrap-Up invite students to think about math anxiety and how they can rephrase those worries into a positive statement about learning math with a Growth Mindset. By posting these positive statements in the classroom, this lesson cultivates a community of Growth Mindset learners and helps students feel supported in attempting new methods or challenging problems.

DIFFERENTIATE INSTRUCTION

Consider using physical base ten blocks for 1.7.2 Guided Instruction to support kinesthetic approaches and help model the vocabulary terms such as “dividend,” “divisor,” “quotient,” and “remainder” with visuals.

In 1.7.4 Try It, prompt students to illustrate the concept of division using visual models first to help them communicate their thinking and help them model the use of the standard division algorithm.



How can this representation of 253 better contribute to number sense?

1.7.1 WARM-UP: WHAT IS MATH ANXIETY?

Component Overview: Display and watch the video as a class, then have students answer the following questions. If technology is not available, students can answer the questions by reading the prompts.

TEACHER MOVES

Collect and Display

Consider having students anonymously write their positive statements about learning math this year on a sticky note or strip of paper, then display these statements in a classroom shared space to promote a Growth Mindset and make students feel safe to make mistakes in the classroom environment.

QUESTIONING

If time permits, consider having students research other prominent mathematicians from around the world and learn how they also exemplified a Growth Mindset.



1.7.1 Warm-Up: What is Math Anxiety?

Have you ever sat down to take a math test and immediately felt your heart beat faster and your palms start to sweat? Sometimes this can happen if you are feeling anxious about math.

1. What are some of your worries about math class this year?

Answers will vary.

Maryam Mirzakhani is an award-winning mathematician who was born in Iran. After being the first woman to win the prestigious Fields Medal, she stated in an interview, "I did poorly in math for a couple of years in middle school; I was just not interested in thinking about it. I can see that without being excited mathematics can look pointless and cold. The beauty of mathematics only shows itself to more patient followers."

2. Remember: the brain is flexible, so your math skills can always grow and develop. Write a positive statement about learning math this year.

Answers will vary.



1.7.2 Guided Instruction: Division of Whole Numbers

Division is

- ☐ repeated addition.
- ☒ repeated subtraction.

This is

because when dividing, we are **partitioning**, or taking out, equal groups from the whole.

1. A division expression is shown.

$$253 \div 12$$

The **divisor** is the size of the groups that are being partitioned.

- a. In the expression above, the divisor is

- ☐ 253.
- ☒ 12.

The **dividend** is the group that is being partitioned.

- b. In the expression above, the dividend is

- ☒ 253.
- ☐ 12.

1.7.2 GUIDED INSTRUCTION: DIVISION OF WHOLE NUMBERS

Component Overview: During this activity, students use base ten block models to investigate how division is partitioning the dividend into equal groups determined by the divisor and how any remaining value is the remainder. Guide students through the scaffolded questions and content in this section as a class to address questions or misconceptions through discussion.

DIFFERENTIATED INSTRUCTION

Consider reading this section aloud to students and pausing at the parts where students choose the correct option. Reading the text aloud provides an auditory entry point that supports students in determining which choices complete the statement. After the class comes to an agreement on the correct choices, ask students to reread the completed text to a partner and label the dividend and the divisor in the expression $253 \div 12$.

ENRICHMENT

Encourage students to think of how they would phrase the key vocabulary in their own words by asking:

- What is another way you can describe the dividend? The divisor?

1.7.2 GUIDED INSTRUCTION: DIVISION OF WHOLE NUMBERS (CONTINUED)

DIFFERENTIATED INSTRUCTION

Consider using base ten blocks to model or have students recreate the model shown for the division problem $253 \div 12$. This kinesthetic approach may engage students to identify how the dividend 253 is being portioned into 12 equal groups using the blocks.

ENRICHMENT

To support making the connection between the expression and the model, prompt student thinking with questions like:

- Which value did you model first?
- Why are you rearranging the model into 12 equal groups?
- Why did you need to regroup the hundreds into tens?
- How do you see the equal groups in your model? Circle the equal groups and put a star next to any blocks that do not fit in the equal groups. What does the star represent?

TEACHER MOVES

On question 2, prompt students to explain the value of the ten-rods in the model and how it relates to the original value of 253 to help them conceptualize why one ten is left over and cannot be partitioned yet. This recognition is essential to understanding how place value and remainder are incorporated in the long division algorithm.

TEACHER MOVES

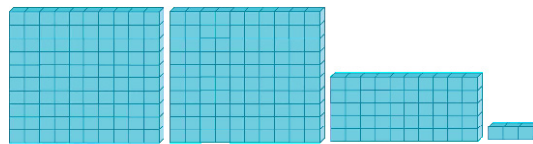
Connect the idea of a remainder in long division to students' understanding of a "remainder" in a different context. For example, ask students what it means if something "remains" or is "left over" and to give examples of when there is a remainder outside of math. This connection between literacy and math vocabulary can help students make sense of division remainders as the remaining amount after equal groups are made.

TEACHER MOVES

Students should begin making sense of partitioning using one of the following stems:

- "Partition 'out' groups of a certain value (or how many in a group)"
- "Determine how many groups of a specific size there might be"

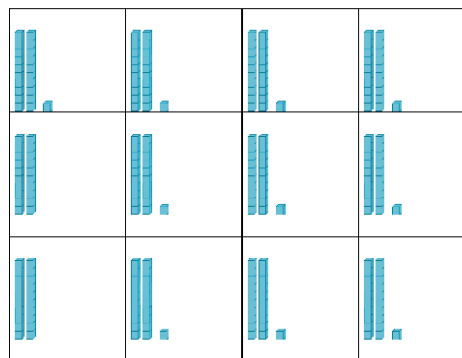
A base-ten model of 253 is shown.



The model should be arranged in 12 equal groups. Since there are not 12 hundreds flats, regroup the hundreds-flats into ten-rods.



2. Partition the ten-rods into 12 equal groups by sketching the ten-rods in the boxes.



3. How many tens are left after partitioning?
1 ten

The remaining ten-rods and the 3 ones should be partitioned by regrouping the ten-rods into ones.

4. Add to your sketch in the boxes the partitioning of the ones.

Shown in question 2

5. How many ones were not able to be partitioned?
1 one

The one-blocks that were not able to be partitioned are the **remainder**. The amount left over can also be written as a fraction, $\frac{\text{remaining one blocks}}{\text{divisor}}$.

6. Write the remainder as a fraction.
 $\frac{1}{12}$

7. What is the value of the base-ten blocks in each group?
21

When there is a remainder, the **quotient** is the value of the base ten-blocks and the remainder written as a fraction.

8. Write the quotient.
 $21\frac{1}{12}$

9. Complete the equation.

$$253 \div 12 = 21\frac{1}{12}$$



1.7.3 Guided Instruction: Partial Quotient

Find $2,743 \div 14$ using the **partial quotient** model. The partial quotient model takes out (subtracts) parts of the total quotient each time. The number of groups that are taken out each time doesn't matter; the quotient will be the same.

1. Complete the table.

Number of Groups of 14	Total Taken Out
1	14
2	28
3	42
4	56
5	70
10	140
20	280
30	420
40	560
50	700
100	1,400
200	2,800
300	4,200
400	5,600
500	7,000

2. What two numbers in the table is 2,743 between?
100 and 200

1.7.3 GUIDED INSTRUCTION: PARTIAL QUOTIENT

Component Overview: Students complete a table with multiples of 14, using these multiples to take out whole groups of 14 from the dividend in the partial quotient model. This activity provides another model of division, moving students from the concrete to the representational and later the abstract long division algorithm.

TEACHER MOVES

Collect and Display

Before completing the table in question 1, have students look for patterns and make observations about what they notice about the values and titles of the two columns. Position students to recognize the connection between division and multiplication and how to use partial quotients to collect responses on the following, then display them in a shared classroom space for students to make connections:

- What relationship do you notice between the number of groups of 14 and the total taken out?
- What operation is the same as “taking out” an amount?
- How do you see multiplication in the table?
- How do you see repeated subtraction in the table?
- How do you think this table could be used to help answer the problem $2,743 \div 14$?

ENRICHMENT

Encourage critical thinking for question 2 by asking:

- Why should we consider which two numbers 2,743 is between? How will that help us solve the problem?

1.7.3 GUIDED INSTRUCTION: PARTIAL QUOTIENT (CONTINUED)

ENRICHMENT

It is critical that students recognize why the smallest group of 14 that could be used is 1,400 even though 2,800 is closer to the dividend 2,743. Position students to recognize the reason behind this choice by asking:

- Notice in the first row we subtracted 1,400. Why is it not possible to subtract 2,800?
- What is the nearest group of 14 that does not go over the dividend value 2,743?
- How can this help you to know what value to subtract first in other partial-quotient problems?

ENRICHMENT

For question 6, elicit student responses that highlight the similarities between the two ways of noting remainders by asking:

- How does the partial product strategy relate to the model before and the expression?
- How does each part of the notation relate to the divisor? The dividend?

3. Use the smaller number of groups of 14 to begin the repeated subtraction. Add rows if necessary, but do not use rows if there are no remaining groups of 14.

2,743	Description
— 1400	100 groups of 14
1,343	Amount that remains
— 700	50 groups of 14
643	Amount that remains
— 560	40 groups of 14
83	Amount that remains
— 70	5 groups of 14
13	Amount that remains
—	_____ groups of 14
	Amount that remains

Answers may vary. Example shown above.

4. How many groups of 14 were subtracted?
195
5. What amount remains?
13
6. Write the quotient using the remainder notation and as a mixed number.
195 R13; $195\frac{13}{14}$



1.7.4 Try It: Division of Whole Numbers

1. Complete the table to find $8,988 \div 25$ using a modified partial quotient model.

Type of Group	Number of Groups of 25	Total Taken Out
ones	1	25
ones	2	50
ones	3	75
ones	4	100
ones	5	125
	etc..	
tens	10	250
tens	20	500
tens	30	750
tens	40	1,000
tens	50	1,250
	etc..	
hundreds	100	2,500
hundreds	200	5,000
hundreds	300	7,500
hundreds	400	10,000
hundreds	500	12,500
	etc..	

2. What two hundreds groups in the table is 8,988 between?
300 and 400

1.7.4 TRY IT: DIVISION OF WHOLE NUMBERS

Component Overview: In this activity, students practice using the partial-quotients method to divide a two-digit divisor into a four-digit dividend and relate the process to the long division standard algorithm.

DIFFERENTIATED INSTRUCTION

If students are having difficulty completing the table, prompt students to think about how the addition of “Number of Groups of 25” in this table is like the model from 1.7.3 Guided Instruction question 1. Position students to generalize that the process is the same, as they are both using the partial-quotient model, but the difference is the values being used for the dividend and divisor (the size of groups).

ENRICHMENT

For question 2, encourage students to think about how place value plays a role in the partial-quotient model by asking:

- Why is it important to look at the two hundreds groups that the answer will be between? What does that tell you about our final quotient?
- How can you use that information to check the reasonableness of your solution?

1.7.4 TRY IT: DIVISION OF WHOLE NUMBERS (CONTINUED)

DIFFERENTIATED INSTRUCTION

Before students begin question 3, invite students to pre-read the information to the right of the problem aloud so they use each of the types of group from the table above to solve this problem only using three rows. Frontloading that this problem should be solved in three steps using the groups of hundreds, tens, and ones as outlined can give students more confidence as they attempt the model.

ENRICHMENT

Facilitate opportunities for the development of number sense through inquiry.

- If we started partitioning with ones, would our number of steps be the same?
- If we started partitioning with ones, would our number of rows be the same?
- Notice your answer has three “hundreds.” Compare this with your answer from question 2. What does that make you think about in terms of checking your answers?

DIFFERENTIATED INSTRUCTION

To support visual learning methods, have students use different colors to divide using the long division method as shown for question 5. Using color to write the values in the quotient and the number of place value groups in the algorithm may visually help students relate how the quotient values connect to the repeated subtraction of groups.

TEACHER MOVES

As partners work, monitor if students are correctly dividing using the long division algorithm and support metacognitive thinking by asking:

- Where do you write the number of groups taken out of the dividend?
- Where do you write the total taken out of the dividend?
- What does the amount that remains represent related to the equal groups of the divisor?
- How could you draw a model with Base Ten Blocks to confirm your answer?
- How can you tell if your solution is reasonable?

3. Use the smaller number of hundreds groups of 25 to begin.

8,988	Description
— 7,500	Number of hundreds groups of 25
1,488	<u>300</u>
	Amount that remains
— 1,250	Number of tens groups of 25
238	<u>50</u>
	Amount that remains
— 225	Number of ones groups of 25
13	<u>9</u>
	Amount that remains

4. What is the quotient of $8,988 \div 25$? Write your answer using a remainder.
359 R13

Another way to show this type of partial quotient is with long division.

5. Use the partial quotient model from above to complete the long division. Notice that the quotient is set up to write the remainder as the partial group of 25.

$$\begin{array}{r} \boxed{3} \boxed{5} \boxed{9} \boxed{13} \\ 25 \overline{) 8,988} \end{array}$$

Number of hundreds groups of 25	— 7500
Amount that remains	<u>1488</u>
Number of tens groups of 25	— 1250
Amount that remains	<u>238</u>
Number of ones groups of 25	— 225
Amount that remains	<u>13</u>

**1.7.5 Your Turn!**

1. Choose a method of division that makes the most sense to you and find each quotient. Show your work and models.

a. $1,276 \div 25$
 $51 R1$

b. $7,839 \div 92$
 $85 R19$

2. The Publix Warehouse was distributing a shipment of bottled water to 36 Publix grocery stores. There was a total of 18,396 packages to distribute. If each store received an equal amount of bottled water, how many packages did they receive?
 511 packages

1.7.5 YOUR TURN!

Component Overview: Students synthesize their understanding of long division using a model, partial quotients, and/or the long division algorithm by individually solving two division problems and one division problem within a real-world context. Students should work independently for practice and to potentially provide data points on individual student learning.

DIFFERENTIATED INSTRUCTION

For question 1, consider having students connect their model to a partial products or long division representation by labelling or drawing arrows to the parts that represent the same division process. This action can help students think with flexibility about different approaches to solving division problems.

ENRICHMENT

For question 2, consider providing an opportunity to think about how the real-world context relates to division by asking:

- What does each value in the problem represent?
- How can you tell what the divisor and dividend are in this problem?
- What number is being partitioned?
- What is the value of each package of bottled water?
- Why is there no remainder in this problem?

1.7.6 WRAP-UP: GROWTH MINDSET

Component Overview: This Wrap-Up provides students with an opportunity to engage in flexible thinking related to a math Growth Mindset. Students revisit ideas from the Warm-Up by again writing a positive statement, but this time it is related to the different division models and strategies learned in this lesson.

TEACHER MOVES

Consider having students compare the progression of their responses from Lesson 5 to this lesson.

QUESTIONING

Both Test Yourself! Practice Tool and EdgeXL can be used as students close Unit 1. Test Yourself! offers a question bank for students to practice questions aligned to the benchmark for this unit (with video solutions), while EdgeXL provides opportunities for customized practice or assessments.



1.7.6 Wrap-Up: Growth Mindset

1. Remember the brain is flexible, so your math skills can always grow and develop. Write a positive statement about what you learned in the lesson.

Answers will vary.

2. Complete the statements.

- I do not know how to do _____
 _____ YET!
- I plan to do _____
 to help my math skills grow and develop.

